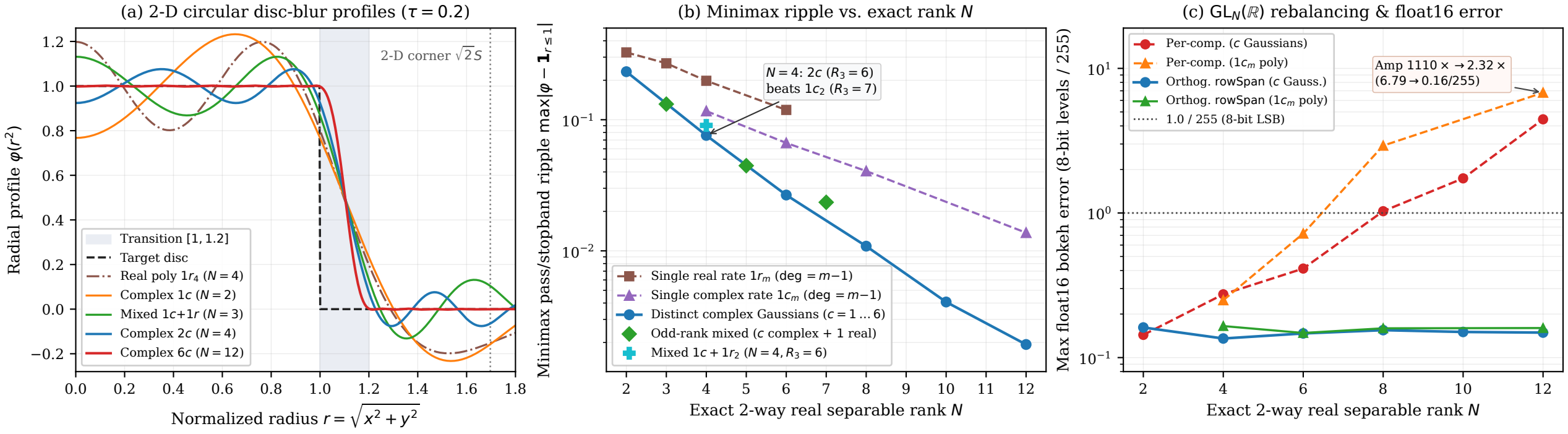

(a) 2-D circular disc-blur profiles ($\tau = 0.2$)
2-D corner $\sqrt{2}S$
Transition [1, 1.2]
Target disc
Real poly $1r_4$ ($N = 4$)
Complex $1c$ ($N = 2$)
Mixed $1c+1r$ ($N = 3$)
Complex $2c$ ($N = 4$)
Complex $6c$ ($N = 12$)
Radial profile $\varphi(r^2)$
Normalized radius $r = \sqrt{x^2 + y^2}$
(b) Minimax ripple vs. exact rank N
$N = 4$: $2c$ ($R_3 = 6$)
beats $1c_2$ ($R_3 = 7$)
Single real rate $1r_m$ (deg $= m-1$)
Single complex rate $1c_m$ (deg $= m-1$)
Distinct complex Gaussians ($c = 1 \ldots 6$)
Odd-rank mixed (c complex + 1 real)
Mixed $1c + 1r_2$ ($N = 4, R_3 = 6$)
Minimax pass/stopband ripple $\max|\varphi - \mathbf{1}_{r \le 1}|$
Exact 2-way real separable rank N
(c) $\mathrm{GL}_N(\mathbb{R})$ rebalancing & float16 error
Per-comp. (c Gaussians)
Per-comp. ($1c_m$ poly)
Orthog. rowSpan (c Gauss.)
Orthog. rowSpan ($1c_m$ poly)
1.0 / 255 (8-bit LSB)
Amp 1110 × → 2.32 ×
(6.79 → 0.16/255)
Max float16 bokeh error (8-bit levels / 255)
Exact 2-way real separable rank N